Ziad Ahmed

+201066634927 | ziaddiab03@gmail.com

Summary

AI Engineer and Data Scientist with a strong foundation in developing and deploying machine learning and deep learning solutions. Skilled in Python, C++, SQL, and MATLAB, with hands-on experience in scalable model development. Proficient in TensorFlow, PyTorch, Keras, and transformer-based models (BERT, GPT, Qwen, Phi). Experienced in preprocessing, feature engineering, and visualization (Pandas, NumPy, Scikit-learn, Plotly). Adept in computer vision (OpenCV) and NLP (NLTK, Hugging Face Transformers). Strong background in full-cycle AI development, including deployment with Docker and Git, and familiar with cloud-based workflows for integrating AI models into production. Earned an M.Sc. in Artificial Intelligence (2019) and a Ph.D. in Computer Science (2024) from Nile University, with research focusing on medical imaging, NLP, and AI-driven cybersecurity. Published research on brain tumor segmentation and 3D reconstruction at the 9th International Undergraduate Research Conference (IUGRC-9). Successfully led the NeuroVista project as Team Leader, coordinating collaboration, task allocation, and advanced model development. Also built an AI-powered intrusion detection system to enhance network and data security. Combines deep technical expertise, academic research, and leadership skills to deliver innovative AI systems that drive business value, improve decision-making, and strengthen digital security.

Technical Skills

Programming Languages: Python, C++, SQL, MATLAB, R, Java

Machine Learning & Deep Learning: TensorFlow, PyTorch, Keras, Scikit-learn, XGBoost, LightGBM, CNNs, RNNs, Transformers, GANs, Reinforcement Learning

Natural Language Processing: NLTK, Spacy, Hugging Face Transformers, BERT, GPT-2, Qwen, Phi, LLaMA, Ollama; expertise in text classification, sentiment analysis, summarization, and fine-tuning LLMs

Computer Vision: OpenCV, YOLO, Faster R-CNN, UNet, nnU-Net, Vision Transformers (ViT)

Data Visualization: Power BI, Tableau, Excel, Plotly, Matplotlib, Seaborn

Data Management & Big Data: SQL Server, MySQL, PostgreSQL, MongoDB, Hadoop, Apache Spark, Apache Kafka; ETL/ELT pipelines

MLOps & Cloud: AWS (SageMaker, S3, EC2), Google Cloud AI Platform, Azure ML, MLflow, Kubeflow, Docker, Kubernetes, Jenkins, CI/CD pipelines

Security & Responsible AI: Adversarial ML, Intrusion Detection Systems, Explainable AI (XAI), Model Fairness & Bias Mitigation

Tools: Git, GitHub, GitLab, Napari, DICOM automation, Jupyter, VS Code, Linux

Soft Skills: Leadership, Teamwork, Strategic Planning, Communication, Problem Solving, Mentorship, Emotional Intelligence

Experience

Software Engineer — Jul 2016 – Aug 2018
TechVision Solutions — *Cairo, Egypt*

- Developed and optimized embedded software modules for automotive systems using C++ and MATLAB.
- Collaborated with cross-functional teams to design and test machine learning models for driver assistance systems.
- Enhanced debugging workflows and improved system performance by 20%.

AI Research Engineer — Sep 2018 – Dec 2021
NTT Data — *Cairo, Egypt*

- Worked on applied AI research projects in natural language processing and computer vision.
- Designed and implemented deep learning pipelines using TensorFlow and PyTorch for document processing and image recognition.
- Published technical papers and presented findings at internal R&D events.

Senior Data Scientist — Jan 2022 – Mar 2024
Orange Business Services — *Cairo, Egypt*

- Led data-driven solutions for telecom analytics, customer churn prediction, and fraud detection.
- Deployed production-ready ML models using Docker and Kubernetes on cloud platforms.

Mentored junior data scientists and collaborated with global teams on scalable AI solutions.

Data Scince instructor Apr 2024 – Oct 2024
IBM — Digital Egypt Pioneers Initiative (DEPI) *Full-Time*

Completed training in Data Analysis, Machine Learning, Data Visualization, and Big Data Technologies.

Applied knowledge in real-world projects, enhancing technical proficiency.

Collaborated with teams and mentors to extract meaningful insights from complex datasets.

Prompt Engineer Jul 2024 – Present
Outlier *Part-Time, Remote*

Designed and engineered prompts to train and evaluate large language models.

Created complex, multi-step tasks and adversarial examples to test robustness.

Collaborated with researchers to ensure quality, diversity, and alignment with objectives.

Projects

Brain Tumor Segmentation & 3D Reconstruction Jul 2025
Deep Learning Research Project *IUGRC-9 Conference*

Developed an end-to-end AI system combining CycleGAN for MRI synthesis, CNN + nnU-Net V2 for tumor segmentation, and Napari for 3D visualization.

Automated DICOM file handling to streamline clinical workflows, overcoming challenges with metadata inconsistencies and segmentation accuracy.

AI-Powered Web Platform for Medical Imaging 2025
Full-Cycle AI Development *Deployed Prototype*

Built a web-based AI platform for automated MRI analysis, integrating deep learning pipelines for preprocessing, segmentation, and visualization.

Implemented scalable deployment using Docker and cloud-based services for clinical usability.

NeuroVista Oct 2024 – Jun 2025
Deep Learning Project *Team Leader*

Led the project, managing collaboration, task allocation, and model development.

Developed CNN-based brain tumor detection models robust to anatomical angles.

Created MRI synthesis model (CycleGAN + ResNet50) to generate T1c from T1 scans.

AI-Powered Intrusion Detection System 2024
Cybersecurity Project *NSL-KDD Dataset*

Developed an intrusion detection model using autoencoders for anomaly detection and supervised learning (Random Forest, XGBoost, CNN) for classification.

Achieved high accuracy in detecting malicious traffic and preventing network intrusions.

Integrated feature engineering and real-time monitoring for scalable security applications.

Sentiment Analysis API Sep 2024 – Dec 2024
Natural Language Processing *Cloud Deployment*

Built a RESTful API for sentiment analysis using Hugging Face transformer models.

Deployed the API on cloud infrastructure for real-world integration.

Diabetes Detection Mar 2023 – Jun 2023
Machine Learning Project

Developed a Random Forest model to predict diabetes risk from medical data.

ACHIEVEMENTS

Published Research Paper Jul 2025
9th International Undergraduate Research Conference (IUGRC-9) *Military Technical College, Cairo*

Published work on brain tumor segmentation and 3D reconstruction.

Developed end-to-end AI-powered web platform using CycleGAN, CNN, and nnU-Net V2.

Automated DICOM handling, resolving metadata and accuracy challenges.

EDUCATION

Nile University Cairo, Egypt
Ph.D. in Computer Science *Graduated: 2024*

Nile University Cairo, Egypt
M.Sc. in Artificial Intelligence *Graduated: 2019*

Modern Academy — Faculty of Computer Science Cairo, Egypt
B.Sc. in Computer Science *Graduated: 2016*

COURSES

IBM Data Science (Jul 2016)

Machine Learning, NTI (Oct 2017)

LANGUAGES

Arabic: Native

English: Proficient